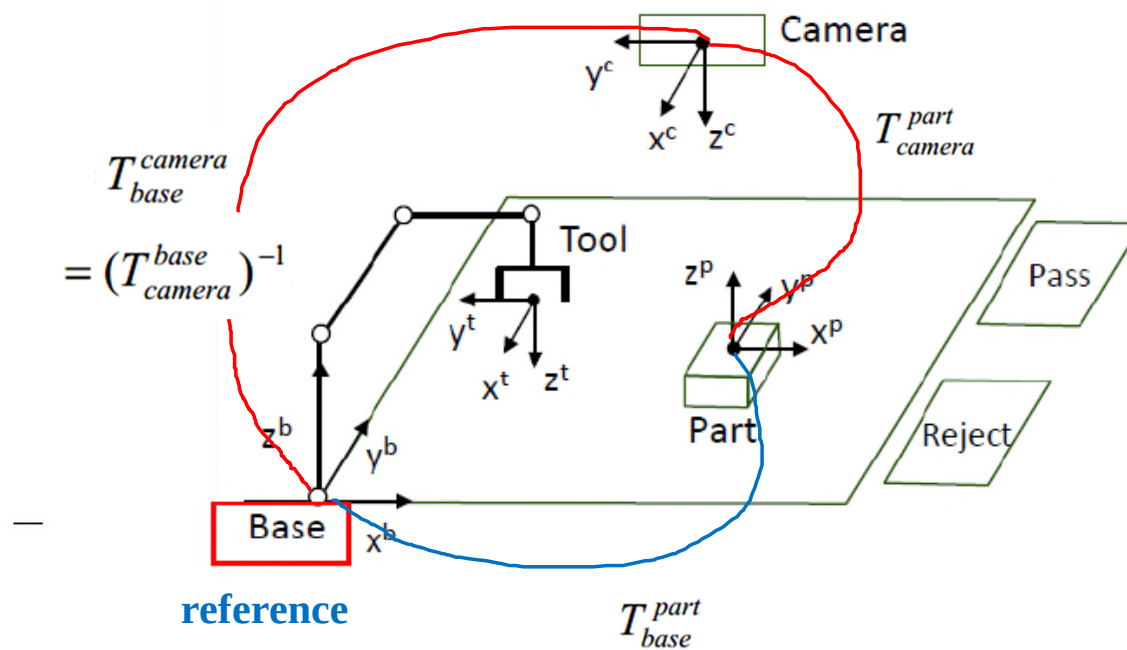
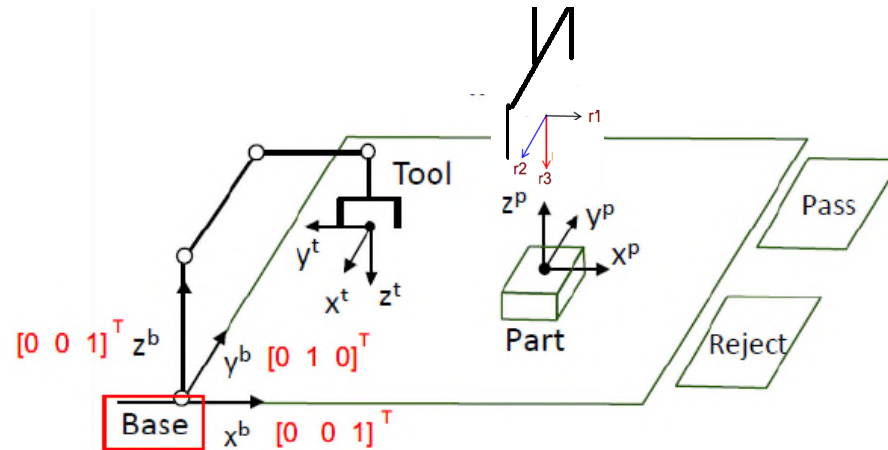


Using homogeneous coordinate transformation matrices, we have

$$T_{base}^{part} = \boxed{T_{base}^{camera}} T_{camera}^{part} = \boxed{(T_{camera}^{base})^{-1}} T_{camera}^{part} = \left( \begin{array}{ccc|c} 1 & 0 & 0 & 30 \\ 0 & 1 & 0 & 15 \\ 0 & 0 & 1 & 1 \\ \hline 0 & 0 & 0 & 1 \end{array} \right)$$

Thus the position of the part relative to the base frame of the robot is  $p = [30, 15, 1]^T$ .





Thus the position of the part relative to the base frame of the robot is  $p = [30, 15, 1]^T$ .

Next suppose that the robot is required to reach down and pick up the part directly from above by grasping it on the front and back faces. In terms of base coordinates, this corresponds to

$$r^3 = -i^3. \quad [0 \ 0 \ -1]^T$$



Next, to grip the object on the front and back faces, a sliding vector is defined as

$$r^2 = \pm i^2. \quad [0 \ \pm 1 \ 0]^T$$

Finally, to complete the right-handed orthonormal coordinate frame,

$$r^1 = \mp i^1. \quad r2 \times r3$$